

# Distributed Systems - 16\_12

The first part of the exam has 6 multiple choices question.

For question 1, remember that when we have  $E_1|p_j$  the  $|$  stands for the local view of the execution.

Ex 1:

The fourth answer is not correct, when they are the same they are indistinguishable, but it's not true that when they're indistinguishable they're the same. The correct answer is 1.

Ex 2:

Here the answer is 3.

Ex 3:

Here there is a problem, it will be fixed on the website.

Ex 4:

It uses symmetric delays. Remember the figure in which we have the server and the client, with the correction factor fixed in the middle between server and client.

Ex 5:

The first answer is also a liveness property, if we take a time  $t$  in which we delivered 5 messages, the property is still not violated.

Ex 6:

"not well formed" means they don't make sense.

If we have "correct", it's not a problem, as the property is well formed for byzantine processes.

So the answer is the second one.

Now, after multiple choices question, there are three open questions.

Ex 7:

The causal broadcast enforce causal ordering. If there is  $m_1 \rightarrow m_2$ , then no process delivers  $m_2$  before  $m_1$ .

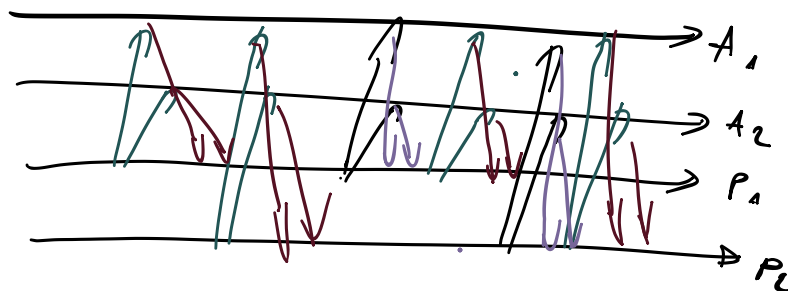
The total order broadcast enforces the exact same ordering on all processes. This order may also not be the causal one.

Now, what are the system assumptions needed? For the causal broadcast, we don't need system assumptions. It can be implemented in asynchronous with any failures. *Remember the easy implementation with the sending of the whole history of a process while sending a message, so that the receiver checks the history and delivers what he still did not deliver.*

As for atomic broadcast, if we solve it in asynchronous system, we also solve consensus (which is not possible by FLP theorem with  $f > 0$ ).

Ex 8:

We need to remember the pattern in which processes kill each other.



In async systems, the example above can run forever, so no liveness.

In esync, when we reach the synchrony period ...

Ex 9:

We have:

- Termination: if a correct process starts an operation, the operation eventually completes.
- LP: the execution is as if any operation immediately executes in a fictional point (linearization point) between the start and the end of the operation.

Next, we have an exercise with an execution.

We can write on the picture.

Now, we have an exercise in which we have to analyze an algorithm.

How do we reason? We ask ourselves what is the impact on the mechanism behind the algorithm.

For example, by having  $from[p]$  rather than from  $s$ , we'll have to check if there is an execution in which we deliver the message twice. This happens if we receive the message both from the relay and the original source.

Let's start by no duplication then, we show an execution:

